Three evidence layers, three different questions

Keep noiseless simulation, finite shot sampling, and QPU measurements separate

NOISELESS VQE

PUCCSD ×3 pilot
90 parameters · 1,374 CX
Median error 0.187 mHa
Best pilot error 0.115 mHa

Separate budget limited run
Best error 0.001520 mHa

Ansatz accuracy

FINITE SHOT AER

PUCCSD ×1 fixed state
16 shots × 276 groups
4,416 nominal shots
Sampling SE 570.414 mHa

Estimate error 789.311 mHa

Finite shot uncertainty

IBM HARDWARE

Four jobs measured HF
Four X gates · zero entanglers
Offsets +50 to +657 mHa
Reported standard errors vary

No optimized VQE state run

Hardware pipeline

The CASCI reference chain is checked independently before comparing VQE energies.

Active space CASCI

−74.860977098519 Ha

Chemical accuracy reference

1.6 mHa from CASCI

ADAPT latest completed cycle

Cycle 11 · 3.473 mHa

A low noiseless error is not the same as a low hardware error. Each reported value belongs to its stated state and estimator.